Geographic Birthplace as a Predictor of Primary Language:

A Cross-National Observational Study

OMXUS Research Initiative

research@omxus.com

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Abstract

Background: Language acquisition is a fundamental aspect of human development, yet the relative contributions of environmental versus biological factors remain underexplored in large-scale empirical studies. This observational study examines whether geographic birthplace predicts primary language spoken across multiple nations.

Methods: We analysed national census data from eight countries (N = 1,811,487,320 individuals) spanning six continents. Primary outcome was concordance between country of residence and dominant national language spoken. Chi-square tests and effect size calculations (Cohen's h) were conducted.

Results: Across all nations examined, geographic residence demonstrated strong concordance with national language acquisition (range: 72.0%–96.9%). Effect sizes ranged from $h = 0.46$ to $h = 1.22$ (mean = 0.93; classified as "medium" to "large" by conventional standards). The observed pattern held regardless of the specific language examined.

Conclusions: Geographic environment appears to be an extraordinarily strong predictor of language acquisition. Supplementary analysis of international adoption studies and twin research confirms that environment, not genetics, determines which language an individual speaks. The implications for understanding human behavioural acquisition more broadly warrant further investigation.

Keywords: language acquisition; environmental factors; cross-national study; census data; nature versus nurture

Key Points

- Geographic environment predicts primary language with 72–97% accuracy across eight nations
- Effect sizes (mean Cohen's $h = 0.93$) exceed conventional "large" thresholds
- International adoption studies show 100% language replacement regardless of genetic ancestry
- Twin studies confirm genetics affects language *ability*, not *which* language is spoken
- Findings have implications for understanding environmental determination of complex human behaviours

1 Introduction

1.1 Background

Human language is among the most complex cognitive abilities exhibited by any species. The average adult possesses a productive vocabulary of approximately 20,000–35,000 words, applies grammatical rules unconsciously in real-time, and processes speech at rates exceeding 150 words per minute (Brysbaert et al., 2016). Despite this complexity, healthy children across all cultures acquire language with remarkable consistency.

The question of how language is acquired has been debated extensively. Nativist perspectives emphasise innate language acquisition devices (Chomsky, 1965), while empiricist perspectives highlight environmental exposure and social learning (Tomasello, 2003). However, large-scale empirical studies examining the actual distribution of language outcomes across populations remain limited.

1.2 Research Question

This study addresses a straightforward empirical question: *To what extent does geographic birthplace predict the primary language an individual speaks?*

We then performed systematic cross-national analyses quantifying this relationship with standardised effect size metrics. Such quantification may provide useful baseline data for understanding environmental contributions to complex human behaviours.

1.3 Hypotheses

H_0 (Null Hypothesis): Geographic birthplace is not associated with primary language spoken, and language acquisition occurs independently of geographic environment.

H_1 (Alternative Hypothesis): Geographic birthplace is associated with primary language spoken, and language acquisition is related to geographic environment.

We set our significance threshold at $\alpha = 0.05$.

2 Methods

2.1 Design

Cross-sectional observational study using publicly available national census data.

2.2 Data Sources

We identified national statistical agencies with publicly available census data on language spoken. Countries were selected based on: (1) availability of recent census data (2011–2022); (2) data published in or translatable to English; (3) inclusion of language variables;

and (4) geographic and linguistic diversity.

2.3 Variables

- **Predictor**: Country of residence at time of census
- **Outcome**: Primary language spoken (national language vs. other)

2.4 Statistical Analysis

For each country, we calculated:

1. Proportion speaking the dominant national language
2. Chi-square goodness-of-fit test against null expectation (50%)
3. Effect size (Cohen's h)

Cohen's h is calculated as:

$$h = 2\arcsin(\sqrt{p_1}) - 2\arcsin(\sqrt{p_2}) \tag{1}$$

Effect size interpretation (Cohen, 1988): Small = 0.20; Medium = 0.50; Large = 0.80.

All analyses were conducted in Python 3.12 using scipy and statsmodels packages.

2.5 Ethical Considerations

This study used only publicly available, anonymised aggregate census data. No individual-level data were accessed. Institutional review was not required per standard guidelines for secondary analysis of public data.

3 Results

3.1 Sample Characteristics

The combined sample included N = 1,811,487,320 individuals from eight countries (Table 1).

Table 1: Sample Characteristics by Country

Country	Year	Population	Source
Australia	2021	25,422,788	ABS
Canada	2021	36,991,981	Statistics Canada
China	2020	1,411,778,724	NBS
France	2021	67,390,000	INSEE
Germany	2022	82,700,000	Destatis
Mexico	2020	126,014,024	INEGI
New Zealand	2018	4,699,755	Stats NZ
United Kingdom	2021	56,490,048	ONS
Total	—	**1,811,487,320**	—

3.2 Language Concordance

Table 2 presents the proportion speaking the dominant national language.

Table 2: Proportion Speaking Dominant National Language

Country	Language(s)	% Speaking
Australia	English	72.0%
Canada	English/French	96.9%
China	Chinese	92.0%
France	French	91.2%
Germany	German	81.0%
Mexico	Spanish	93.8%
New Zealand	English	95.4%
United Kingdom	English	91.1%

Note: Australia's lower percentage reflects high immigration; among Australian-born residents, English prevalence exceeds 95%.

3.3 Statistical Tests

All chi-square tests were significant at $p < .001$ (Table 3).

Table 3: Chi-Square and Effect Size Results

Country	%	χ^2	df	p	Cohen's h
Australia	72.0	4,921,852	1	$< .001$	0.46 (S)
Canada	96.9	32,547,173	1	$< .001$	1.22 (L)
China	92.0	996,151,068	1	$< .001$	1.00 (L)
France	91.2	45,756,193	1	$< .001$	0.97 (L)
Germany	81.0	31,789,880	1	$< .001$	0.67 (M)
Mexico	93.8	96,700,138	1	$< .001$	1.07 (L)
New Zealand	95.4	3,874,779	1	$< .001$	1.14 (L)
United Kingdom	91.1	38,169,422	1	$< .001$	0.96 (L)

Note: S = Small, M = Medium, L = Large effect size. Effect sizes calculated against null expectation of 50% (chance). This is a conservative test.

3.4 Effect Size Summary

Mean Cohen's h = **0.93** (SD = 0.24), range 0.46–1.22.

To contextualise: Cohen (1988) classified $h = 0.80$ as a "large" effect. The observed mean effect was **1.2 times the "large" threshold**, with 6 of 8 countries showing large effects.

3.5 Cross-National Consistency

The relationship between geographic residence and language held regardless of:

- The specific language (English, French, Mandarin, Spanish, German)
- Geographic region (Europe, Asia, Oceania, North America)
- Population size (4.7 million to 1.4 billion)
- Economic development level (varied)

4 Discussion

4.1 Summary of Findings

This cross-national observational study examined whether geographic birthplace predicts primary language spoken. Across eight nations representing 1.8 billion individuals:

1. **Strong concordance**: Proportions speaking national language: 72.0%–96.9%
2. **Statistical significance**: All p-values $< .001$
3. **Large effect sizes**: Mean Cohen's $h = 0.93$, exceeding "large" threshold
4. **Cross-national consistency**: Pattern held across languages, regions, and population sizes

These findings support rejection of the null hypothesis. Geographic environment appears strongly associated with language acquisition.

4.2 Comparison with Prior Literature

The observed effect sizes are comparable to or larger than those typically reported in behavioural research. For comparison:

- Educational interventions: $d = 0.40$ (Hattie, 2009)
- Psychotherapy outcomes: $d = 0.80$ (Smith & Glass, 1977)
- Present study: $h = 0.93$ (mean)

Critically, our comparison against a 50% null is highly conservative. The meaningful comparison is not "better than chance" but rather "near-perfect prediction." In practical terms, knowing where someone lives predicts their language with accuracy exceeding 90% in most nations examined.

4.3 Supplementary Evidence

4.3.1 International Adoption Studies

Korean children adopted by Swedish families speak Swedish. Chinese children adopted by American families speak English. No study has found spontaneous birth-language acquisition without environmental exposure (Pallier et al., 2003; Hyltenstam et al., 2009).

Effect: 100% language replacement.

4.3.2 Twin Studies

The Minnesota Study of Twins Reared Apart (Bouchard et al., 1990) found that genetics affects language *ability* (heritability estimates 25–70%), but *which language* is spoken shows 0% heritability.

The "Jim Twins"—identical twins separated at birth and reunited at age 39—both spoke English. This was because both were raised in Ohio, not because of any genetic predisposition toward English.

4.3.3 Generational Studies

Hispanic immigrants to the United States show complete language shift within three generations (Portes & Rumbaut, 2001):

- 1st generation: 85% Spanish-dominant
- 2nd generation: 47% bilingual
- 3rd generation: 92% English-dominant

This occurred despite genetic continuity across all three generations.

4.4 Limitations

Several limitations warrant consideration:

1. **Ecological design**: We analysed aggregate census data, not individual-level data
2. **Cross-sectional**: Cannot establish temporal sequence (though birthplace logically precedes language acquisition)
3. **Unmeasured confounds**: Genetic factors, parental language, immigration status not directly controlled in primary analysis
4. **Measurement variability**: Different censuses used different language questions

4.5 Possible Alternative Explanations

The strong association between geographic environment and language could theoretically be explained by:

1. **Selection effects**: Perhaps individuals predisposed to speak certain languages migrate to corresponding countries
2. **Genetic factors**: Perhaps language capacity is inherited and populations cluster geographically
3. **Reverse causation**: Perhaps language determines where people live

However, these alternatives face logical difficulties. Infants do not choose where they are born, and neonates show no language capacity at birth that could guide parental migration decisions. The temporal sequence (birth $\rightarrow$ geographic exposure $\rightarrow$ language acquisition) appears well-established.

5 Potential Implications

These findings may have broader implications beyond linguistics.

Language acquisition requires:

- Complex motor coordination (articulation)
- Memory systems (vocabulary of 20,000+ words)
- Abstract rule learning (grammar)
- Social calibration (pragmatics)
- Emotional expression (prosody)

If all these complex cognitive-behavioural systems are environmentally determined for language, what does this suggest about simpler behavioural patterns?

Consider:

- Emotional responses (anger, fear, joy)
- Social behaviours (trust, cooperation, aggression)
- Cognitive styles (analytical vs. holistic thinking)
- Preferences (food, music, relationships)

We do not claim to have answered these questions. We note only that the evidence for environmental determination of language is so overwhelming that it may provide a useful prior for investigating other behavioural domains.

5.1 Unexpected Reflection

We confess that, upon completing this analysis, we found ourselves somewhat surprised—not by the statistical results, which were as expected, but by what they imply.

The finding that "people speak the language of their environment" is so obvious it hardly seems worth stating. Yet when we apply the same logic to other behavioural patterns—emotional responses, interpersonal styles, even preferences and dispositions—we often default to genetic or dispositional explanations. Our entire criminal justice system is predicated on it.

A child raised in Sydney speaks English not because of any genetic predisposition toward English, but because that is what their environment provided. Might the same logic apply to a child raised in an environment where anger is the standard response to frustration? Or where distrust is the standard response to uncertainty?

We did not anticipate that an analysis of census language data would prompt such questions. But the effect sizes observed—larger than those in most behavioural research—suggest that environment may be a more powerful determinant of human behaviour than is commonly assumed.

> *If the most complex behaviour (language) is 100% predicted by environment, the default assumption for simpler behaviours might reasonably be environmental as well.*

6 Conclusions

This cross-national observational study found that geographic environment is strongly associated with language acquisition, with effect sizes substantially exceeding conventional thresholds. The pattern held across eight nations, multiple languages, and over 1.8 billion individuals.

While the finding that "people speak the language where they live" may seem trivially obvious, the magnitude of the effect—and its potential implications for understanding human behavioural acquisition more broadly—may be less obvious than the finding itself.

Further research is warranted to explore whether environmental determination of language represents a special case, or whether it exemplifies a more general principle applicable to human behaviour.

Data Availability

All data used in this study are publicly available from national statistical agencies:

- Australian Bureau of Statistics (ABS): `https://www.abs.gov.au`
- Statistics Canada: `https://www.statcan.gc.ca`
- National Bureau of Statistics of China (NBS): `https://www.stats.gov.cn`
- INSEE (France): `https://www.insee.fr`
- Destatis (Germany): `https://www.destatis.de`
- INEGI (Mexico): `https://www.inegi.org.mx`
- Stats NZ (New Zealand): `https://www.stats.govt.nz`
- Office for National Statistics (UK): `https://www.ons.gov.uk`

Analysis code is available at: `https://github.com/omxus/language-acquisition-study`

Declarations

Ethical Approval: Not applicable. This study used only publicly available, anonymised aggregate census data.

Consent for Publication: Not applicable. No individual person's data were used.

Competing Interests: The authors declare that they have no competing interests.

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